



2005

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 17, 21 and 25

Total marks – 100

Section I Pages 2–27

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 29–43

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

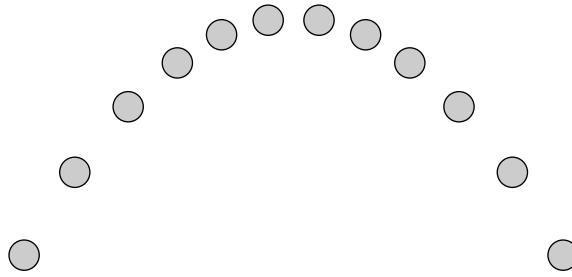
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1 A ball thrown in the air traces a path as shown below.



Which of the following statements is true?

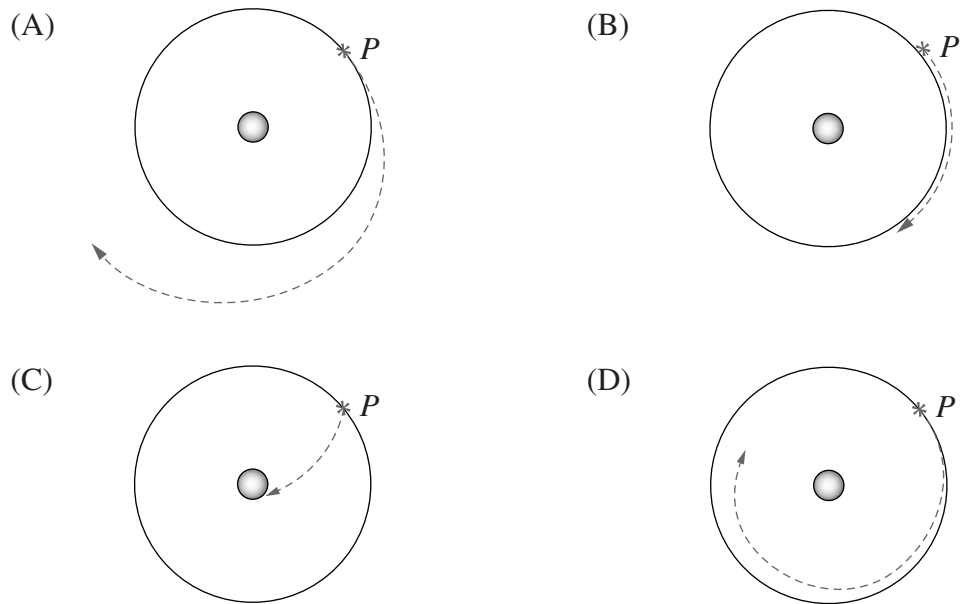
- (A) The velocity of the ball keeps changing.
 - (B) The acceleration of the ball keeps changing.
 - (C) The velocity of the ball at the top of its motion is zero.
 - (D) The acceleration of the ball at the top of its motion is zero.
- 2 Why would a satellite in low orbit around Earth eventually fall to Earth?
- (A) It is not in a geostationary orbit.
 - (B) Gravity is too strong at low orbits.
 - (C) The sun's solar wind pushes it out of orbit.
 - (D) The upper atmosphere gradually slows it down.
- 3 The initial velocity required by a space probe to just escape the gravitational pull of a planet is called *escape velocity*.

Which of the following quantities does NOT affect the magnitude of the escape velocity?

- (A) Mass of the planet
- (B) Mass of the space probe
- (C) Radius of the planet
- (D) Universal gravitational constant

- 4 A space probe, P , is in a stable orbit around a small, distant planet. The probe fires a forward-facing rocket that reduces its orbital speed by half.

Which of the following best illustrates the subsequent motion of the probe?

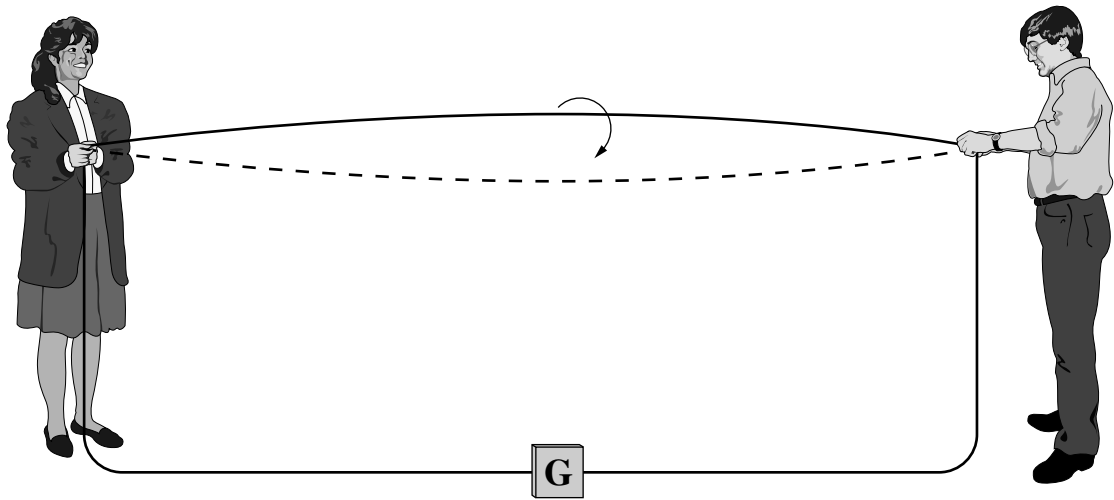


- 5 Napoleon attacked Moscow in 1812 with his cannon firing a shot at an elevation angle of 40° . Napoleon then decided to fire a second shot at the same speed but at an elevation angle of 50° .

Which of the following observations would Napoleon expect to be true about the second shot when compared with the first?

- (A) Longer range
- (B) Shorter range
- (C) Longer time of flight
- (D) Shorter time of flight

- 6 In a particular experiment a long length of copper wire of very low resistance is rotated by two students. The ends of the wire are connected to a galvanometer, G, and a current is detected.

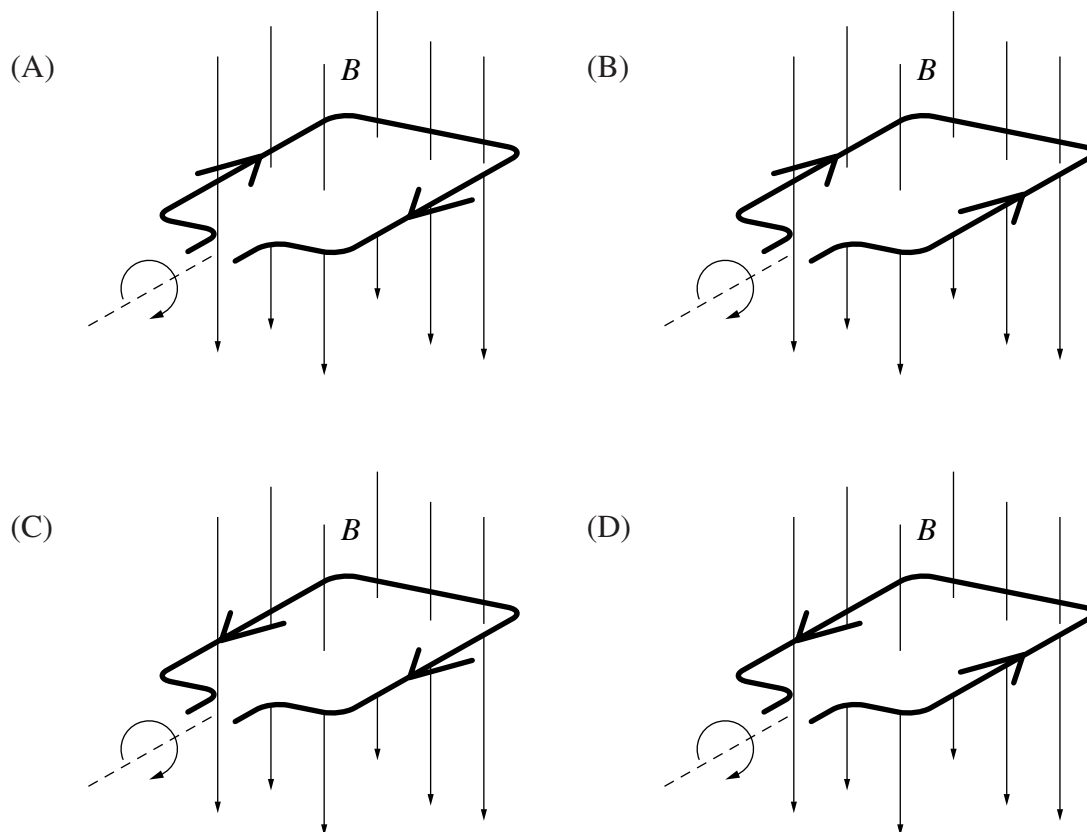


Which of the following is LEAST likely to affect the amount of current produced?

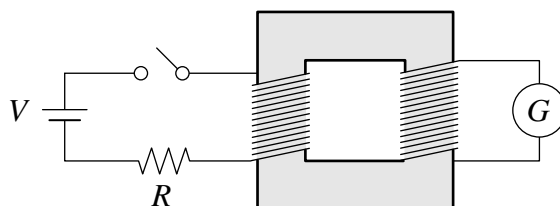
- (A) The length of the rotating wire
- (B) The thickness of the rotating wire
- (C) The speed with which the wire is rotated
- (D) Whether the wire is oriented north-south or east-west

- 7 A single-turn coil of wire is placed in a uniform magnetic field B at right angles to the plane of the coil as shown in the diagrams. The coil is then rotated in a clockwise direction as shown.

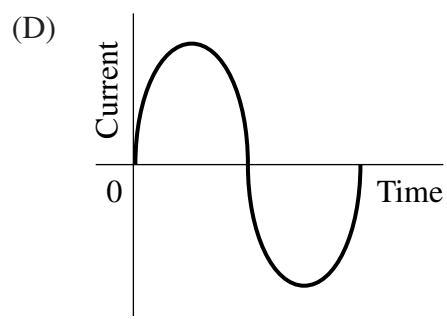
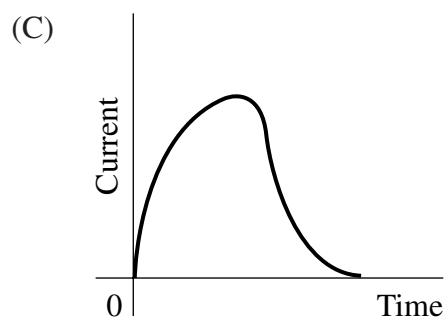
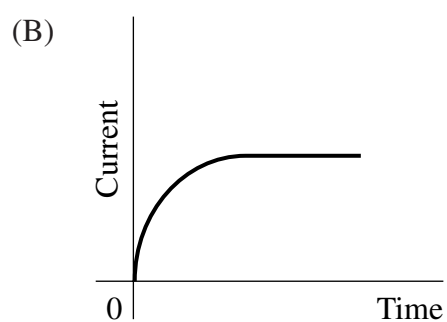
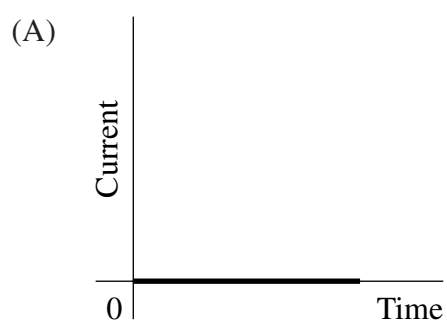
Which of the following shows the direction of current flow in the coil as it begins to rotate?



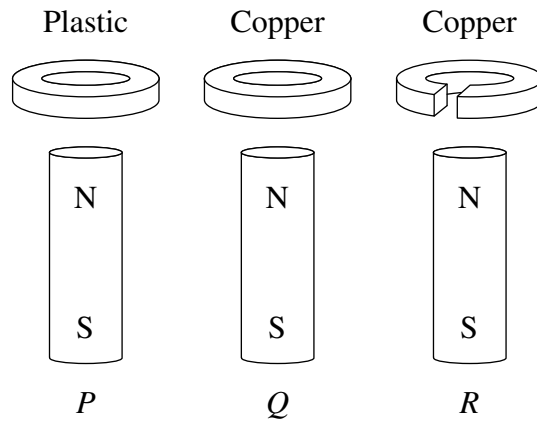
- 8 The primary coil of a transformer is connected to a battery, a resistor and a switch. The secondary coil is connected to a galvanometer.



Which of the following graphs best shows the current flow in the galvanometer when the switch is closed?



- 9 Three rings are dropped at the same time over identical magnets as shown below.

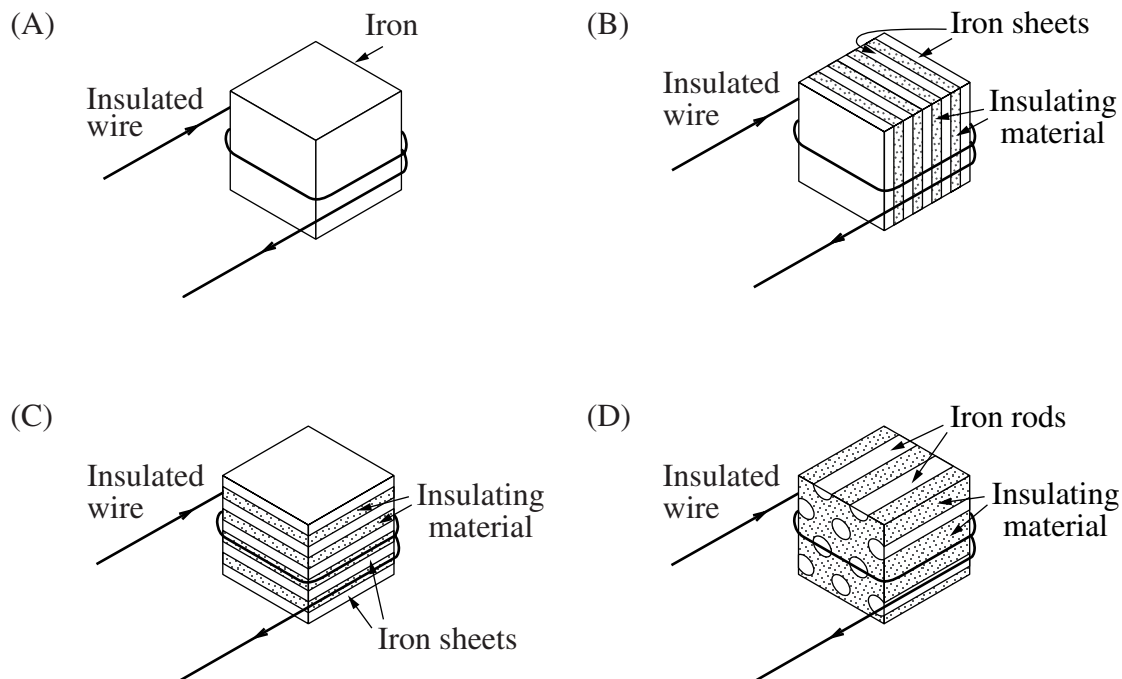


Which of the following describes the order in which the rings *P*, *Q* and *R* reach the bottom of the magnets?

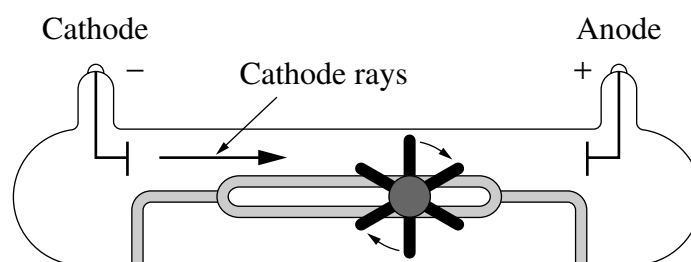
- (A) They arrive in the order *P*, *Q*, *R*.
- (B) They arrive in the order *P*, *R*, *Q*.
- (C) Rings *P* and *R* arrive simultaneously, followed by *Q*.
- (D) Rings *Q* and *R* arrive simultaneously, followed by *P*.

- 10 A transformer is to be designed so that it is efficient, with heating by eddy currents minimised. The designer has some iron and insulating material available to build the transformer core. The windings are to be made with insulated copper wire.

Which of the following designs minimises the energy losses in the core?



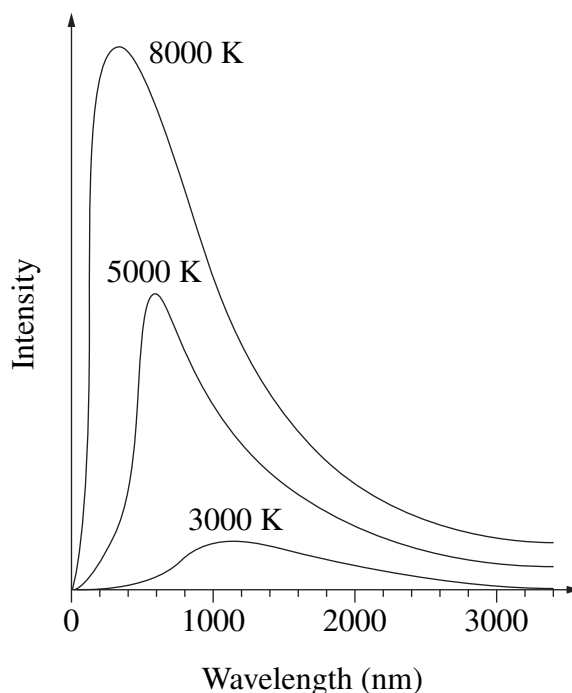
- 11 The discharge tube shown below contains a rotating paddle wheel that is free to move. The tube's electrodes are connected to a high-voltage source.



Which of the following statements about cathode rays does this apparatus provide evidence for?

- (A) Cathode rays travel in straight lines.
- (B) Cathode rays are particles that have momentum.
- (C) Cathode rays can only be produced in vacuum tubes.
- (D) Cathode rays are waves of high frequency and short wavelength.

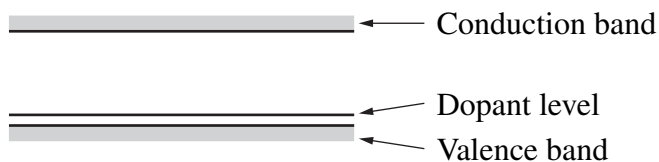
- 12 The family of curves below shows the relationship between the intensity of black body radiation and its wavelength for various Kelvin temperatures.



This diagram has been adapted from Figure 2.18 in Physics Concepts and Applications, VCE Units 1&2 by Harding et al, Macmillan Education Australia, 1997. Reproduced by permission of Macmillan Education Australia.

Who was the first to correctly explain this relationship?

- (A) Planck, in 1900, when he suggested energy at the atomic level was quantised
 - (B) Einstein, in 1905, when he suggested light was a stream of particles called photons
 - (C) Rutherford, in 1911, when he suggested the nuclear model of the atom
 - (D) Bohr, in 1913, when he suggested electrons exist in stationary states
- 13 A doped silicon semiconductor has the following energy-level diagram.



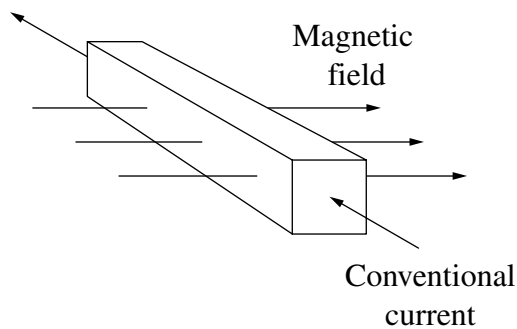
What element was most likely used to dope the silicon?

- (A) Boron
- (B) Germanium
- (C) Phosphorus
- (D) Sulfur

- 14 An FM radio station transmits at a frequency of 102.8 MHz.

What is the energy, in joules, of each photon emitted by the transmitter?

- (A) 6.446×10^{-42}
(B) 6.812×10^{-26}
(C) 2.918
(D) 3.084×10^{16}
- 15 A current is passed along a square semiconductor rod as shown. Half of the current is carried by electrons and half by holes. A magnetic field is then applied to the rod at right angles to its axis.



Which of the following correctly describes the movement of the electrons and holes in the rod when the magnetic field is applied?

- (A) They speed up.
(B) They slow down.
(C) They move to the same side of the rod.
(D) They move to opposite sides of the rod.

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (5 marks)

From nearest to furthest, the four satellite moons of Jupiter first observed by Galileo in the year 1610 are called Io, Europa, Ganymede and Callisto. For the first three moons, the orbital period T of each is exactly twice the period of the one orbiting immediately inside it. That is,

$$T_{\text{Europa}} = 2 \times T_{\text{Io}}$$

$$T_{\text{Ganymede}} = 2 \times T_{\text{Europa}}$$

The mass of Jupiter is 1.90×10^{27} kg, and the orbital radius of Io is 421 600 km.

- (a) Use Kepler's Law of Periods to calculate Ganymede's orbital radius.

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- (b) Calculate Ganymede's orbital speed.

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Question 17 (6 marks)

Einstein's 1905 theory of special relativity made several predictions that could not be verified for many years.

- (a) State ONE such prediction. **1**

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- (b) Describe an experiment to test this prediction. **2**

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- (c) Explain how technological advances since 1905 have made it possible to carry out this experiment. **3**

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Question 18 (4 marks)

The idea of a universal aether was first proposed to explain the transmission of light through space. Michelson and Morley attempted to measure the speed of Earth through the aether.

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Evaluate the impact of the result of the Michelson and Morley experiment on scientific thinking.

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Please turn over

Question 19 (4 marks)

In 1970 NASA launched Apollo 13, their third mission planned to land humans on the Moon. Half-way to the Moon a huge explosion crippled the spacecraft. The only way home for the astronauts was to fly around the back of the Moon and then fire the rocket engine to take the craft out of lunar orbit and put it into an Earth-bound trajectory.

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At the completion of the rocket engine burn, mission leader Jim Lovell was heard to say, 'We just put Isaac Newton in the driver's seat'.

Given that the spacecraft returned safely to Earth, justify Jim Lovell's statement.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 20 (6 marks)

In your course you had to gather information to explain how induction is used in certain applications.

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With reference to TWO applications, describe how you assessed the reliability of information you found.

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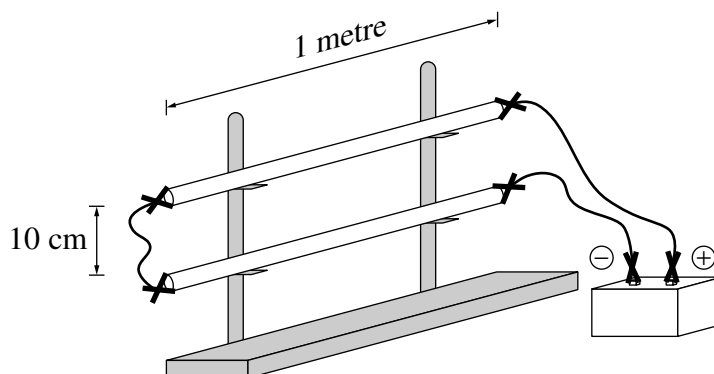
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Question 21 (6 marks)

Two thin metal tubes one metre long were supported in a vertical wooden rack as shown in the diagram.



The two ends were connected together, then the other two ends were connected briefly to a car battery as shown in the diagram. It was observed that one of the tubes jumped upward as the connection was made.

- (a) Explain why only one tube jumped upward. 2

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- (b) Each tube has a mass of 1×10^{-2} kg, and the tubes lie on the rack 10 cm apart. 3

What minimum current flows when one tube jumps?

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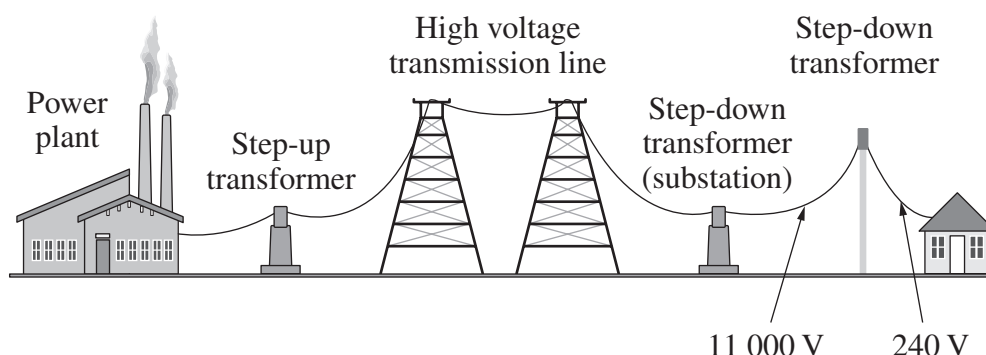
- (c) What is the implication of this result for power distribution networks? 1

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Question 22 (5 marks)

A schematic diagram of a system to supply electricity to a house is shown below.



J D Cutnell & K W Johnson, 2001, *Physics*, 5th edn. Reprinted with permission of John Wiley & Sons, Inc.

The step-down transformer in the substation has a turns ratio of 30 : 1.

- (a) What is the voltage carried by the high voltage transmission line? **1**

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- (b) Identify the causes of the two main energy losses in the transmission of electricity between the power plant and the house. **2**

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- (c) Explain how the application of superconductivity could minimise energy loss in the system. **2**

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Question 23 (3 marks)

Explain how an understanding of black body radiation changed the direction of scientific thinking in the early twentieth century.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 24 (4 marks)

Using labelled diagrams and text, explain how superconductivity occurs according to the BCS theory.

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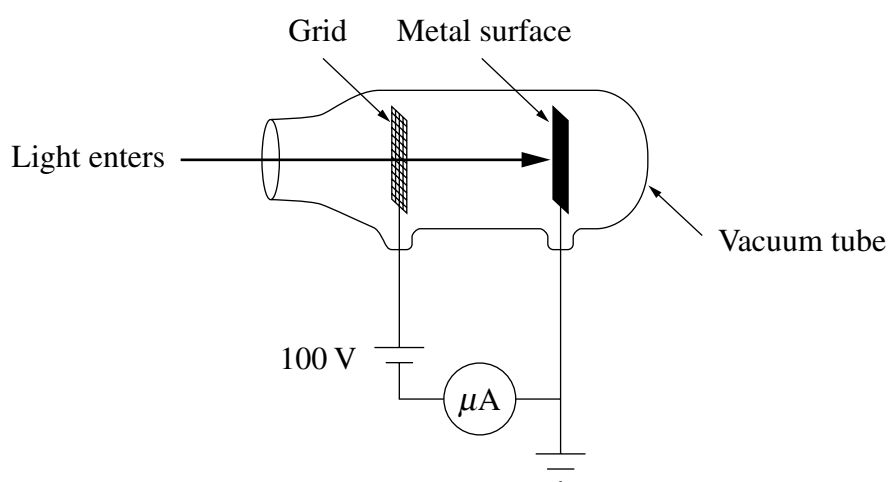
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Question 25 (6 marks)

A student conducts an experiment using a photoelectric cell as shown in the diagram.



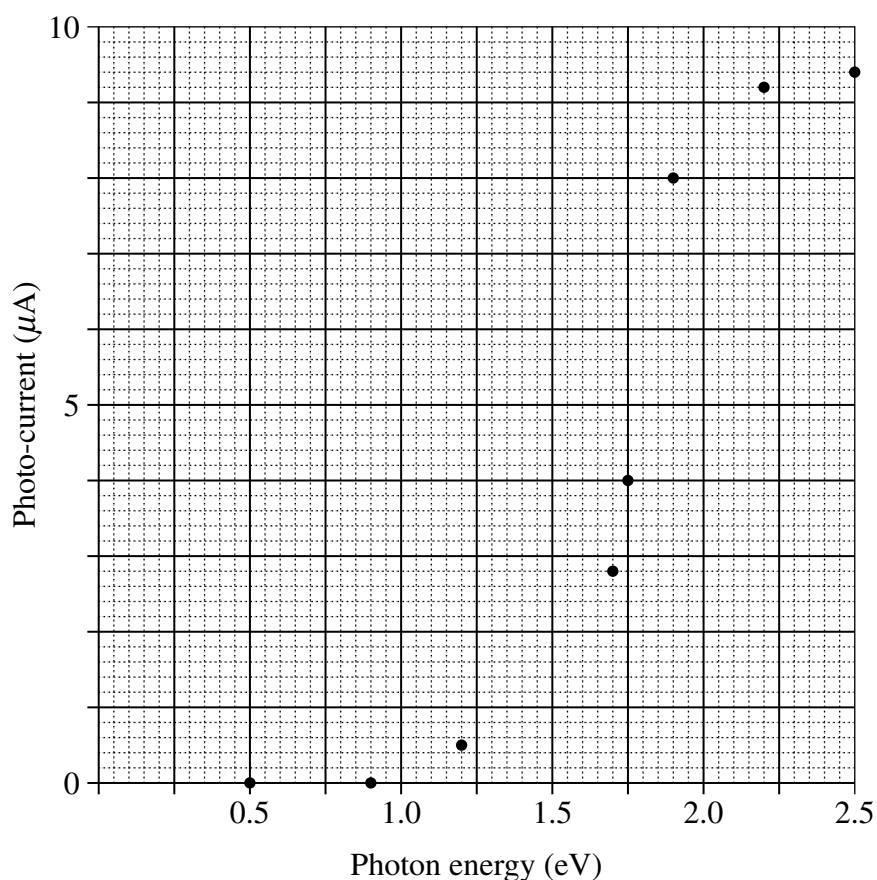
Light is shone through a grid onto a metal surface. The metal is at earth potential and the grid is at 100 V, so that any electrons emitted from the surface produce a current in the external circuit.

The student shines light sources of different photon energies onto the metal surface and records the current flowing for each. The light sources are adjusted so that their intensities are equal. The results are recorded in the table and shown on the graph.

<i>Photon energy (eV)</i>	<i>Photo-current (μA)</i>
0.50	0
0.90	0
1.20	0.5
1.70	2.8
1.75	4.0
1.90	8.0
2.20	9.2
2.50	9.4

Question 25 continues on page 23

Question 25 (continued)



- (a) On the grid provided, draw the straight line of best fit in the region where the photo-current varies greatest with photon energy. 1

- (b) From the line drawn on your graph, estimate the minimum energy (work function) for photoelectric emission. 1

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- (c) The experiment is repeated, but the intensities of the light sources are doubled. Predict the results of this new experiment by drawing a second line on the graph. 2

- (d) Justify the line you have drawn in part (c). 2

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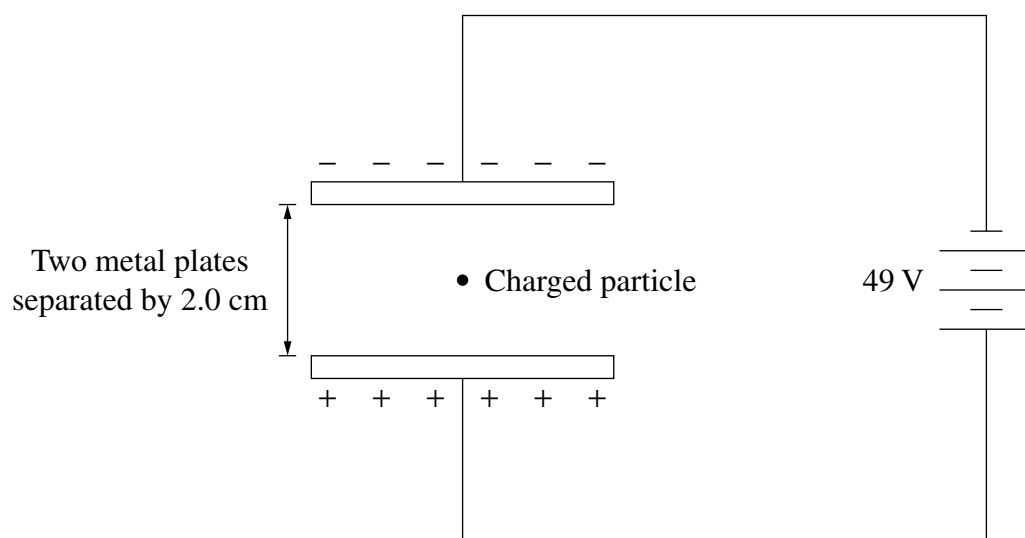
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End of Question 25

Question 26 (5 marks)

The diagram shows two parallel horizontal metal plates connected to a DC source of electricity. Suspended between the plates is a charged particle of mass 9.6×10^{-6} kg.



- (a) Using conventional symbols, draw the electric field between the metal plates on the diagram above. 1

- (b) Determine the magnitude of the electric field between the plates. 1

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- (c) Determine the sign and magnitude of the charge on the particle if it is suspended motionless between the plates. 3

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Centre Number

Section I – Part B (continued)

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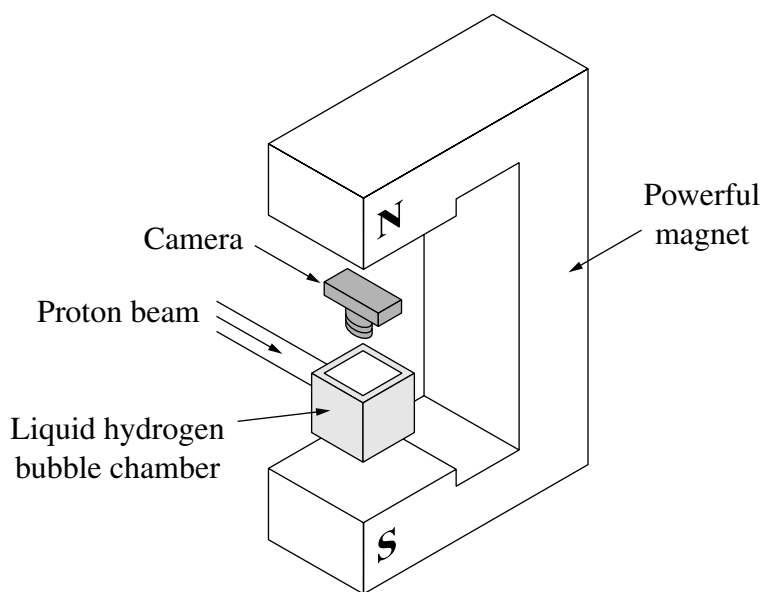
Question 27 (6 marks)

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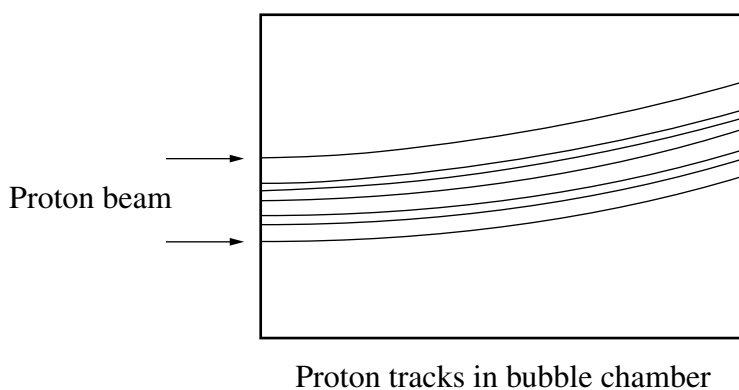
Question 27 (6 marks)

Bubble chambers are used in conjunction with particle accelerators to photographically record the tracks of fast-moving charged particles. An intense magnetic field is applied at right angles to the path of the particles to deflect them according to their charge and momentum.

The diagram shows a beam of protons travelling horizontally at $6.0 \times 10^7 \text{ m s}^{-1}$ and entering a liquid hydrogen bubble chamber in a vertical magnetic field of 1.82 T.



Examination of the photograph taken by the camera, as sketched below, shows that the protons were deflected along a circular path of radius 0.350 metres.



Question 27 continues on page 27

Question 27 (continued)

- (a) Derive an expression for the momentum of a proton from the forces it experiences in this experiment. **2**

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- (b) Calculate the mass of a proton in the bubble chamber. **2**

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- (c) Calculate the rest mass of a proton found from this experiment. **2**

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End of Question 27

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Physics

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Geophysics	30–31
Question 29 Medical Physics	32–35
Question 30 Astrophysics	36–38
Question 31 From Quanta to Quarks	39–41
Question 32 The Age of Silicon	42–43

Question 28 — Geophysics (25 marks)

- (a) During your study of geophysics you investigated the radiation reflected from various surfaces.

(i) Identify the equipment you used to obtain your results. **2**

(ii) Describe the use of reflected radiation in obtaining information about Earth from a distance. **2**

- (b) On 6 December 2004 a meteor exploded in the atmosphere above northern NSW. The blast was detected by sensitive microphones in Hobart at 5.25 am (AEST) and in Tennant Creek at 6.12 am (AEST).

AEST = Australian Eastern Standard Time



(i) If Hobart is 1320 km from the explosion, how far is Tennant Creek from the explosion? **2**

(ii) Similar microphones have detected volcanic explosions such as the Mount St Helens (USA) volcanic explosion in 1980. **4**

Identify another geophysical technique and explain how it is used to locate a volcanic explosion.

Question 28 continues on page 31

Question 28 (continued)

- (c) The development of technologies that increased our understanding of Earth's magnetic field led to the acceptance of the principle of plate tectonics. **7**

Evaluate this statement.

- (d) The diagram below shows the deflection of a plumb-bob near a large mountain range. The diagram exaggerates the amount of deflection.



- (i) Explain why the plumb-bob is deflected towards the mountain range. **2**
- (ii) The observed deflection towards the mountain range is not as great as predicted due to the mountains alone. **3**

What is the implication of this for plate tectonics?

- (iii) Describe how Jean Richer used a pendulum to show that Earth is not spherical. **3**

End of Question 28

Question 29 — Medical Physics (25 marks)

- (a) (i) The images show a person's heart before and after a medical procedure. **2**



**Abnormal heart
before procedure**

**Heart
after procedure**

Describe how radioactive isotopes have been used to identify the abnormality and confirm its correction.

- (ii) The table provides examples of some radioactive isotopes and their properties. **2**

<i>Radioactive source</i>	<i>Radiation emitted</i>	<i>Half-life</i>
^{11}C	Gamma	20.30 minutes
^{99}Tc	Gamma	6.02 hours
^{201}Tl	Gamma	3.05 days
^{131}I	Gamma	8.04 days
^{137}Cs	Alpha	30.17 years
^{238}U	Alpha	4.47×10^9 years

Which radioactive isotope from the table would most likely be used to investigate the abnormality shown in the image above? Justify your choice.

Question 29 continues on page 33

Question 29 (continued)

- (b) (i) The acoustic impedance of fat is $1.38 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$.

The acoustic impedance of bone is $7.80 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$.

What percentage of the incident intensity of an ultrasound wave is reflected as it crosses from fat into bone?

- (ii) Compare the physics involved in producing X-ray images with that used for endoscopies.
- (c) The images demonstrate advances in the use of ultrasound as a tool in medical diagnosis.



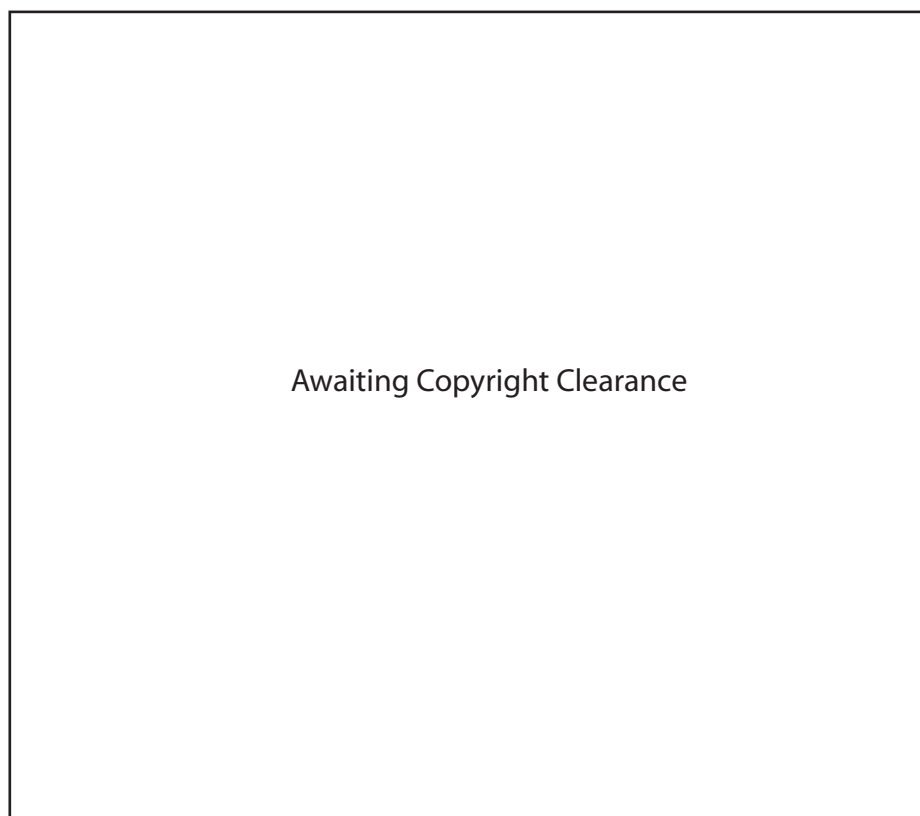
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Describe advances in technology that have enabled the improvements shown in these images, and discuss current issues that have arisen from these advances.

Question 29 continues on page 34

Question 29 (continued)

- (d) (i) The following diagram shows the constituent parts of an MRI system. **2**



State the functions of the superconducting magnet assembly and the radio frequency (RF) coils in the MRI system.

Question 29 continues on page 35

Question 29 (continued)

- (ii) The use of MRI may be improved by the introduction of gadolinium into the body. 2

**T_1 curves for tissues *A* and *B*
without gadolinium in the body**

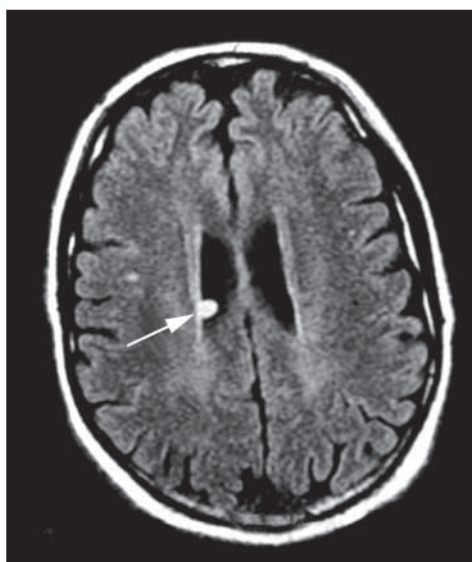
**T_1 curves for tissues *A* and *B*
with gadolinium in the body**

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Explain why gadolinium has been introduced.

- (iii) The arrow indicates an abnormality that has been detected in one hemisphere of the brain.

MRI brain scan



Identify the advantages of MRI over a CAT scan in detecting this abnormality.

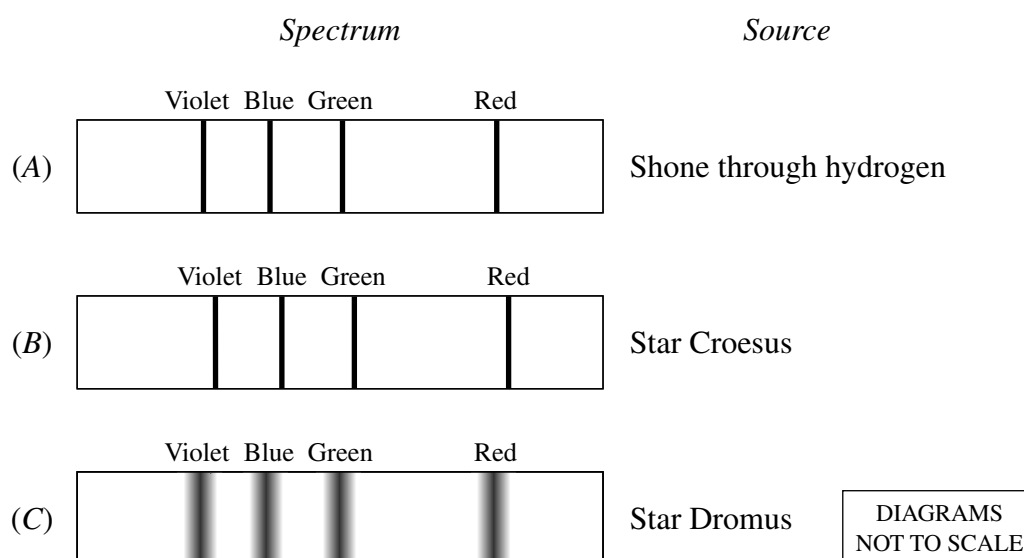
End of Question 29

Question 30 — Astrophysics (25 marks)

- (a) Part A of the figure shows the absorption spectrum of light, produced by an incandescent filament, after it has been shone through a quantity of hydrogen gas.

Also shown in the figure are the spectra obtained from two stars, Star Croesus in part B and Star Dromus in part C.

The dark lines are absorption bands in A, B and C.

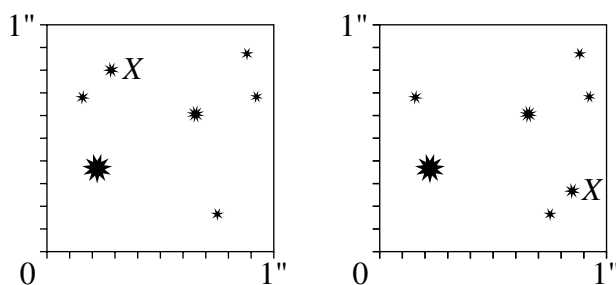


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| (i) For each star, Croesus and Dromus, identify the principal way in which its spectrum differs from the spectrum shown in part A of the figure. | 2 |
| (ii) For each star, Croesus and Dromus, state what its spectrum tells us about the motion of that star. | 2 |

Question 30 continues on page 37

Question 30 (continued)

- (b) (i) Photographs taken of a one arcsecond by one arcsecond sector of the night sky show a group of fixed stars. Scales have been added to the photographs. One star appears to change position, swinging backward and forward over a period of one year. Two photographic negatives taken when the star was at the furthest ends of its apparent travel are shown. The star is marked *X*. 2



Calculate the distance of the star *X* from Earth.

- (ii) When viewed through a telescope, the star Alpha Centauri is seen to be three stars close together. Two of them are the very bright Alpha Centauri A and the very faint Proxima Centauri. These stars are 1.3 pc from Earth. Their magnitudes are given in the table below. 4

<i>Star</i>	<i>Absolute magnitude</i>
Alpha Centauri A	+ 4.33
Proxima Centauri	+14.93

What is the ratio of their apparent brightnesses?

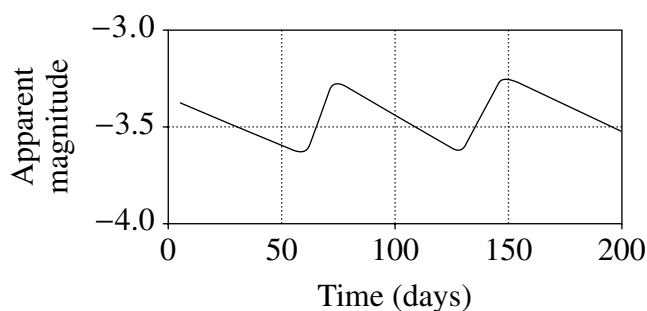
- (c) The Hertsprung–Russell (or H–R) diagram relates the magnitude or brightness of stars to their spectral classes or temperatures. 7

Describe the technological advances that have made it possible to add astrophysical data to the H–R diagram, and explain how this data contributes to our understanding of stellar evolution.

Question 30 continues on page 38

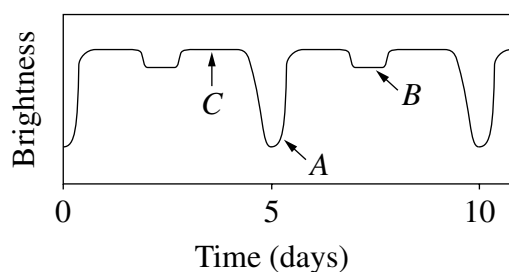
Question 30 (continued)

- (d) (i) The graph shows the apparent magnitude of a supergiant star recorded over a period of time. The star is identified as a Type I Cepheid variable. 2



Explain how the period of oscillations in apparent magnitude may be used to determine the distance of the star.

The graph shows the brightness of a star system recorded over a period of time. The star system is identified as a binary pair, and measurements show them to be 5.0×10^{10} m apart. One star is known to have four times the mass of the other.



- (ii) Explain what causes each of the features *A*, *B* and *C* labelled on the graph. 3
- (iii) Determine the mass of the star with the smaller mass. 3

End of Question 30

Question 31 — From Quanta to Quarks (25 marks)

- (a) During your study of From Quanta to Quarks you either performed a first-hand investigation, or you gathered information to observe nuclear radiation using a Wilson cloud chamber, or similar detection device.

Below is a true-size photograph in this type of device showing the tracks made by α -particles.

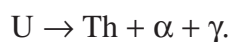


- (i) Explain the appearance of these tracks in terms of properties of α -particles. **2**
- (ii) Name another type of nuclear radiation, and describe differences in the tracks it would make. **2**

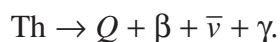
Question 31 continues on page 40

Question 31 (continued)

- (b) Naturally occurring uranium-238 spontaneously disintegrates according to the equation



The thorium radionuclide undergoes further decay according to the equation



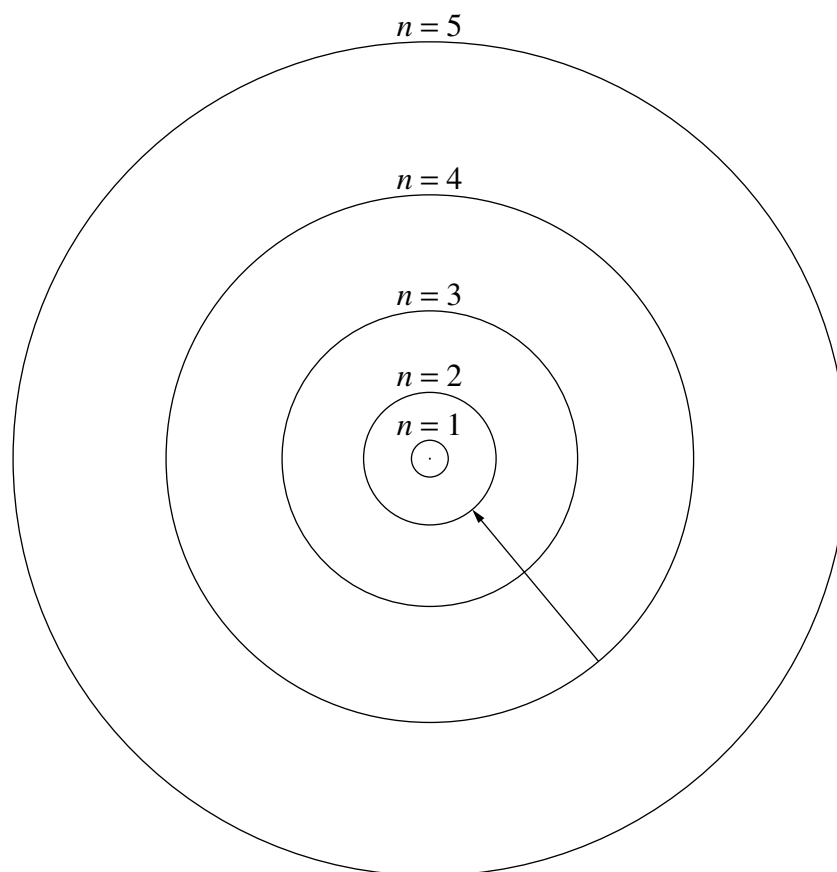
- | | |
|---|----------|
| (i) Identify the mass number of the thorium radionuclide. | 1 |
| (ii) Identify the nuclide Q , stating its mass number. | 2 |
| (iii) Describe Wolfgang Pauli's contribution to Enrico Fermi's explanation of beta decay. | 3 |
-
- | | |
|--|----------|
| (c) An understanding of the nucleus led to the Manhattan Project, which was based in laboratories in Los Alamos between 1942 and 1945. | 7 |
|--|----------|

Describe the technologies developed from this project, and assess the significance to science and society of their applications.

Question 31 continues on page 41

Question 31 (continued)

- (d) The diagram below shows the first five circular Bohr orbits or ‘stationary states’ for the electron orbiting the nucleus of the hydrogen atom.



- | | |
|---|----------|
| (i) For the electron transition shown on the diagram, calculate the wavelength of the emitted photon. | 2 |
| (ii) State de Broglie's hypothesis, and calculate the wavelength of the electron in the first stationary state if its speed is $2.188 \times 10^6 \text{ m s}^{-1}$. | 3 |
| (iii) Describe how de Broglie's hypothesis extended the work of Bohr in explaining the stability of electron orbits in the hydrogen atom. | 3 |

End of Question 31

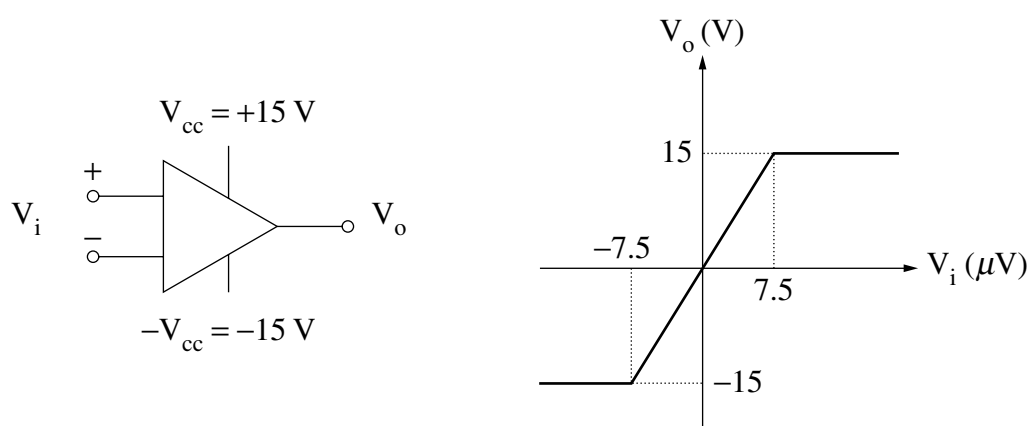
Question 32 — The Age of Silicon (25 marks)

- (a) (i) Write down the truth table for the logical expression 2

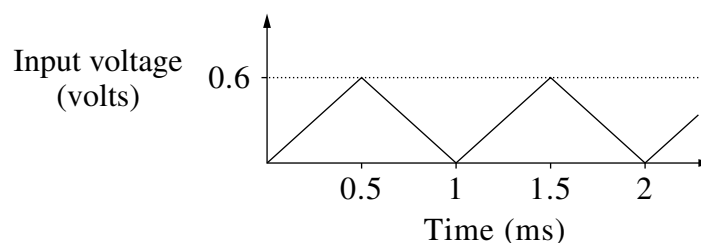
$$C = \text{NOT } (A \text{ AND } B)$$

- (ii) Describe the function of a half-adder, and draw a circuit diagram to show how logic gates can be used in combination to make a half-adder. 2

- (b) An operational amplifier has the transfer characteristic shown.



- (i) Design an amplifier with a gain of -50 using the above operational amplifier, and describe the difference between open-loop and closed-loop gain in your amplifier. 4
- (ii) In your writing booklet, sketch the output voltage of your amplifier as a function of time if the input voltage is a triangular wave as shown. 2



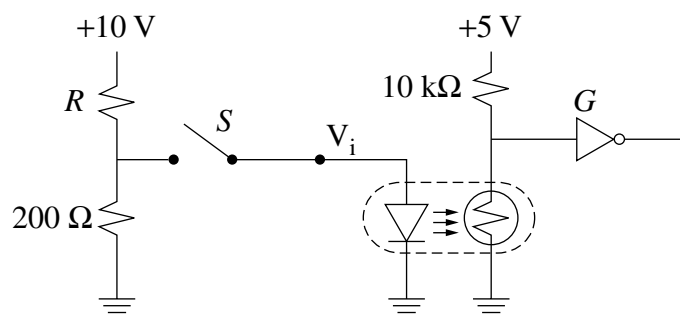
Question 32 continues on page 43

Question 32 (continued)

- (c) Over the last ten years the ability to acquire, store and manipulate digital images has increased dramatically. 7

Describe the advances in semiconductor technology responsible for this increased ability, and explain how such changes have led to new consumer electronics applications.

- (d) (i) Distinguish between input and output transducers, giving an example of each. 2
- (ii) The circuit below uses an optical isolator (comprising a LED and LDR) to electronically isolate a switch S from a digital gate G . 3



When switch S is closed, a current of 20 mA flows through the LED and the voltage V_i is 1.6 V.

Determine the resistance R .

- (iii) The following table shows a variation of the resistance of the LDR as a function of LED current. 3

<i>LED current</i>	<i>LDR resistance</i>
30 mA	190 Ω
20 mA	290 Ω
10 mA	600 Ω
1 mA	20 k Ω
< 0.1 mA	> 1 M Ω

Show, using calculations, how the digital output of the gate G , either '1' or '0', depends on whether the switch is open or closed.

End of paper

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \quad \text{therefore} \quad a_{\text{av}} = \frac{v-u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium											
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	Atomic Number		Symbol of element		Atomic Weight		Name of element		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon			
		79 Au 197.0 Gold		Au		197.0 Gold		Gold		13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon			
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium																						
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.41 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton						
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.94 Molybdenum	43 Tc [97.91] Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon						
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanides		72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.8 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po [209.0] Polonium	85 At [210.0] Astatine	86 Rn [222.0] Radon					
87 Fr [223.0] Francium	88 Ra [226.0] Radium	Actinides		104 Rf [261.1] Rutherfordium	105 Db [262.1] Dubnium	106 Sg [266.1] Seaborgium	107 Bh [264.1] Bohrium	108 Hs [277] Hassium	109 Mt [268] Meitnerium	110 Ds [271] Darmstadtium	111 Rg [272] Roentgenium												

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [144.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [244.1] Plutonium	95 Am [243.1] Americium	96 Cm [247.1] Curium	97 Bk [247.1] Berkelium	98 Cf [251.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
---------------------------------	------------------------------	-----------------------------------	-----------------------------	----------------------------------	----------------------------------	----------------------------------	-------------------------------	----------------------------------	------------------------------------	------------------------------------	---------------------------------	-------------------------------------	----------------------------------	------------------------------------

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.

**2005 HSC Notes from
the Marking Centre
Physics**

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2005 HSC NOTES FROM THE MARKING CENTRE

PHYSICS

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It provides comments with regard to responses to the 2005 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

This document should be read along with the relevant syllabus, the 2005 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

General Comments

In 2005, 9315 candidates attempted the Physics examination. The most popular electives were From Quanta to Quarks (44%) and Astrophysics (27%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

In 2005, at least one question in Section 1 Part B focused on the mandatory skills content in Module 9.1. Candidates who had actively planned and performed practical experiences clearly demonstrated a deeper knowledge and understanding of the content described in this module. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with the course.

Overall, the candidate's responses were appropriate and indicated a level of understanding of Physics concepts appropriate for most HSC candidates. Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce clear and concise responses. This may include the use of dot points, diagrams and/or tables, and avoids internal contradictions. This is particularly so in holistic questions which need to be logical and well-structured.

Better responses indicate that candidates are following the instructions provided on the examination paper. In these responses, candidates:

- show all working for numerical questions and express their answers with the correct number of significant figures
- think carefully about the units to be used and the quantities to be substituted into formulae
- do not repeat the question as part of the response

- look at the structure of the whole question and note that in some questions the parts follow from each other ie responses in part (a) lead to the required response in part (b) etc
- use appropriate equipment, for example, pencils and a ruler to draw diagrams and graphs. (A clear plastic ruler would aid candidates to plot points that are further from the axes and rule straight lines of best fit.)

The option question is divided into a number of parts: candidates should clearly label each part of the question when writing in their answer booklets. In part (c), many candidates wrote a lot of information that was not relevant to the question. Some responses show evidence of rote learning an anticipated answer based on a single source. These did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Some candidates are responding to more than one option question: candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	A
2	D
3	B
4	D
5	C
6	B
7	A
8	C

Question	Correct Response
9	C
10	B
11	B
12	A
13	A
14	B
15	C

Part B

Specific Comments

Question 16

- (a) In better responses, candidates demonstrated recognition of the required formula provided in the Formulae Sheet and provided correct substitution of values to determine the period of Io and hence the period of Ganymede. The responses included the use of Kepler's formula again to determine the orbital radius of Ganymede.

In the best responses, candidates recognised that $\frac{GM}{4\pi^2}$ was a constant and determined the radius using ratios.

Weaker responses involved substitution errors or the use of incorrect units.

- (b) In better responses, candidates recognised that centripetal force and gravitational force had to be equated to derive an equation for orbital velocity and substituted the correct values and units.

The weaker responses included substitution errors or used 'r' values in kilometres rather than metres.

Question 17

- (a) Better responses to this question began with a simple statement such as 'time dilation'.
- (b) Better responses described a valid experiment for the statement in part (a) eg one involving two atomic clocks, one on a high speed jet and the other earth-bound.
- (c) Better responses provided a clear explanation of how the nominated technological advances allowed the experiment to be successfully completed.

Weaker responses did not link parts (a), (b) and (c).

Question 18

Better responses discussed the impact on the scientific community of the result of the Michelson-Morley experiment and its implication for the existence of the aether. This shifted scientific thinking from classical theory to relativity.

Question 19

In better responses, candidates were able to describe and relate two or more of Newton's Laws of Motion in words and/or equations to the return trip of Apollo 13, ie the period after leaving the lunar orbit.

In weaker responses, candidates incorrectly applied the 'slingshot effect' to the motion of the rocket around the moon.

Question 20

The best responses made a judgement about the reliability of information gathered both from first-hand data and/or second-hand data. Many candidates understood the general principles of assessing reliability of information but had difficulty relating these principles to the identified applications of induction.

Question 21

- (a) The better responses clearly identified a repulsion force existed between the two tubes and went on to explain why one moved. Responses that incorrectly identified attraction between the tubes were still able to explain why only one tube jumped. A common poorer response was to relate the brief connection of the circuit to an induced back emf and attempt to use this to explain the upward jump.
- (b) The best responses calculated the mathematical relationship between weight and electromagnetic force between the two tubes. Candidates are reminded that the use of SI units is essential to express a correct value for current.
- (c) Good responses indicated factors that influence forces between closely placed conductors carrying large amounts of current.

Question 22

- (a) Most responses correctly identified the formula and included a correct calculation of the voltage in the high tension wires.
- (b) In a majority of responses the major causes of energy losses were identified as resistance in the wires and eddy currents in the transformers.
- (c) This section was generally well answered with the majority of responses indicating that superconductors have zero resistance. However, a significant number of these responses did not include an explanation of how this minimised the energy losses.

Question 23

There was a broad range of responses to this question, most indicated that the emission and absorption of black body radiation as quanta was understood to be important.

A number of responses included a graph of intensity versus wavelength (or frequency) in an attempt to describe the need for Planck's hypothesis; if done correctly this proved helpful. In some responses, the labels on the axes were often omitted or confused.

Better responses indicated why Planck's hypothesis was needed and linked this to a change in scientific thinking.

Question 24

A majority of responses identified that the BCS theory involved Cooper Pairs of electrons. Better responses drew clear diagrams and clearly labelled the sections of the diagrams. These responses indicated a clear understanding between the drop of resistance with temperature. Clearly structured answers showing point by point description of the theory and showed that candidates understood the terminology used to describe the BCS theory, for example ‘phonons’ are integral to the explanation.

Question 25

This question assessed Module 9.1 skills content. A large number of responses indicated difficulties in interpreting the graph and relating it to the photoelectric effect.

- (a) & (b) The best responses clearly had a thin, ruled line of best fit that extended through the data points to the x-axis so that the minimum energy could accurately be determined. Poorer responses did not identify the region of greatest variation in the data and did not extend the line of best fit to the x-axis.
- (c) Better responses drew a straight line of steeper gradient from the same x-intercept. Poor responses included a curved line.
- (d) Better responses described the link between the increased steepness of the second line and an increase in the number of electrons released from the metal surface due to a doubling of the number of photons in the incident light. They identified that it would have the same x-intercept because the minimum energy would not change.

Question 26

- (a) The best responses showed clearly on the diagram a uniform electric field between both plates.
- (b) The best responses correctly identified and substituted into the correct electric field equation showing the conversion of centimetres into metres.
- (c) Better responses clearly identified the balance between the downward weight force and the upward force of the electric field and clearly stated that the charge was positive. Weaker responses indicated confusion between electric and magnetic fields.

Question 27

- (a) The better responses derived an expression for the momentum of a proton in the bubble chamber by equating the centripetal force to the magnetic force.
- (b) The best responses included correct substitution into the expression found by equating centripetal force and magnetic force.
- (c) The better responses were those that recognised that the mass obtained in part (b) was a relativistic mass and substituted accordingly into the Lorentz transformation.

Section II – Options

Question 28

- (a) (i) The majority of responses named the correct equipment.
 - (ii) Poorer responses confused emitted Infra Red with reflected Infra Red.
- (b) (i) Better responses used the value for the speed of sound from the Data Sheet to calculate the correct distance.
 - (ii) Good responses named a geophysical technique and explained how to use the equipment and data.
- (c) Better responses identified the technologies that lead to the acceptance of the principle of plate tectonics and made a judgement based on the evidence they presented.
- (d) (i) Better responses correctly identified the gravitational attraction and related it to the deflection of the plumb-bob.
 - (ii) This part of the question was generally well done.
 - (iii) A majority of candidates demonstrated an understanding of using a pendulum to find variations in 'g' and how this related to the shape of the earth.

Question 29

- (a) Many responses demonstrated a sound knowledge of the process of obtaining an image of parts of the body using a radioactive isotope. In better responses there was a link to the specified procedure and a justification the choice of radioisotope in terms of the effect of the radiation produced and the appropriateness of the half-life.
- (b) (i) Candidates were required to select the appropriate equation and substitute correctly into it to determine the percentage of an incident ray reflected at a boundary. The majority of candidates were able satisfy both these requirements and the question was generally well answered.
 - (ii) Better responses focused on and identified several of the actual physical principles involved, and used the cited principles to make a clear comparison between the two imaging technologies.
- (c) Better responses showed a thorough understanding of how ultrasound scans were obtained and addressed the 'advances in technology' leading to the improvements in scans. Weaker responses either only identified advances in technology or identified or outlined issues.
Better responses addressed both parts of this question.

- (d) (i) Better responses correctly stated that the superconducting magnet assembly produced a very strong magnetic field which aligned the protons in the body and also stated the function of the radio frequency coils as pulsing RF waves into the body in order to resonate with aligned protons which subsequently re-emit radio waves to again be detected by the radio frequency coils.
- (ii) Better responses correctly identified that the gadolinium affects the relaxation time of tissue A more than tissue B and consequently enhances clarity or definition of the resultant scan. The majority of responses correctly suggested that improvement of the image was the reason for the introduction of the gadolinium.
- (iii) Better responses identified that the most significant advantage of MRI over CAT is its ability to distinguish between soft tissues and also referenced this technique to the brain and to the abnormality shown in the MRI brain scan image. They also identified that MRI is a safer technique and that MRI is capable of discerning function as opposed to mere structure.

Question 30

- (a) (i) The better responses used correct terminology.
- (ii) The best responses provided information about the motion of both stars. For the star Dromus many candidates wrote only about density rather than rotation. Weaker responses incorrectly stated Dromus was orbiting another star.
- (b) (i) Good responses involved the halving of the measurement from the graph when substituting. Many candidates were confused by the two dimensional angular displacement.
- (ii) The best responses correctly calculated the apparent magnitude using the brightness ratio formula. Good responses involved correct substitution into the distance modulus formula using log to the base 10 (log) and not the natural log (ln).
- (c) The better responses provided characteristics and features of a number of technological advances and of the evolution of stars. These responses clearly related the technological advances to putting new data onto the H-R diagram, and how this leads to an understanding of the evolution of stars.
In weaker responses the technologies were merely identified or outlined
- (d) (i) The better responses provided relevant information in point form.
- (ii) This question was generally well answered, providing reasons for the causes of A, B and C. Most candidates identified the system as an eclipsing binary system. Weaker responses explained the shape of the binary curve due to differences in mass or size of the stars, not brightness.
- (iii) Better responses showed full working for calculations. This included stating the correct formula and giving a data summary, as well as converting the information into the correct units.

Question 31

- (a) Most responses indicated some knowledge of cloud chamber observations.
- (b) Better responses showed an understanding of radioactive decay equations. They were able to link Pauli's ideas to the work of Fermi. Weaker responses were unable to identify Fermi's ideas on beta decay.
- (c) The best responses often included diagrams and/or tables.

The better responses were well balanced answers providing features and characteristics of two or more nuclear technologies developed from the Manhattan project and making a judgement about the significance of their impact on society.

- (d) This question was generally well answered by a majority of candidates.

In parts (i) and (ii) the majority of responses identified the correct equations and made the correct substitutions.

- (iii) In better responses candidates demonstrated an understanding that only standing waves of integral wavelength supported Bohr's postulate by not losing energy.

Question 32

- (a)
 - (i) Most candidates were able to provide the required responses.
 - (ii) Better responses gave a circuit, a truth-table and a description.
- (b)
 - (i) Many responses displayed just a basic op-amp. Few responses gave calculations of resistances or labelled their design of the circuit.
 - (ii) Most responses identified correctly negative output. Few responses gave correct scaling and identified clipping.
- (c) Better responses addressed the key issues of both 'digital images' and 'last ten years', citing relevant examples.
- (d)
 - (i) A majority of responses provided appropriate examples to distinguish between the two transducers.
 - (ii) Few responses identified the key steps of calculating the total current through the resistor.
 - (iii) This part was generally well answered.

Physics

2005 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.2.2.2.1	H9
2	1	9.2.2.2.11	H7
3	1	9.2.2.2.3	H9, H13
4	1	9.2.2.2.10	H9
5	1	9.2.2.2.1	H6
6	1	9.3.2.3.2	H9
7	1	9.3.2.3.1	H9, H14.3 (c)
8	1	9.3.4.3.1	H7, H12.3 (c)
9	1	9.3.2.2.7, 9.3.2.3.4	H7
10	1	9.3.4.3.3	H7
11	1	9.4.1.3.2	H10
12	1	9.4.2.2.3	H10
13	1	9.4.3.2.6	H10
14	1	9.4.2.2.6, 9.4.2.3.4	H7, H12.4 (b)
15	1	9.4.3.2.3, 9.3.1.2.5	H9
Section I Part B			
16	5	9.2.2.2.10, 9.2.2.3.5	H6, H12, H13
17	6	9.2.4.3.4, 9.2.4.2.10	H2, H6, H14
18	4	9.2.4.1	H1, H2, H6, H8, H14
19	4	9.2.3.2.3, 9.2.3.2.2	H6, H9, H14
20	6	9.3.2.3.3, 9.3.2.3.4	H7, H9, H12
21	6	9.3.1.2.2, 9.3.1.3.1, 9.3.3.2.4	H4, H9, H12, H13
22	5	9.3.4.3.2, 9.3.4.2.3, 9.3.3.2.4, 9.4.4.2.5	H4, H7, H9, H12

Question	Marks	Content	Syllabus outcomes
23	3	9.4.2.2.3, 9.4.2.2.4	H1, H10, H14
24	4	9.4.4.2.6	H9, H13, H14
25	6	9.4.2.2.5, 9.4.2.2.6	H7, H10, H12, H13, H14
26	5	9.4.1.3.3, 9.4.1.2.7	H9, H12, H13
27	6	9.4.1.2.5, 9.2.4.3.5, 9.2.2.3.4	H6, H9, H12, H13
Section II			
Question 28 — Geophysics			
28 (a) (i)	2	9.5.2.3.1	H7, H11
28 (a) (ii)	2	9.5.2.2.1	H7, H14
28 (b) (i)	2	9.5.3.3.2	H8, H12
28 (b) (ii)	4	9.5.1.2.2, 9.5.3.2.4	H8, H14
28 (c)	7	9.5.1, 9.5.4.1	H3, H14
28 (d) (i)	2	9.5.2.2.4	H6, H9, H14
28 (d) (ii)	3	9.5.2.2.4	H6, H9, H14
28 (d) (iii)	3	9.5.2.3.2	H6, H9, H14
Section II			
Question 29 — Medical Physics			
29 (a) (i)	2	9.6.3.3.2	H10, H12, H14
29 (a) (ii)	2	9.6.3.2.1	H10, H12, H14
29 (b) (i)	2	9.6.1.3.5	H8, H12, H13
29 (b) (ii)	4	9.6.2.2.1, 9.6.2.2.5	H8, H10
29 (c)	7	9.6.1	H3, H5, H14
29 (d) (i)	2	9.6.4.3.3	H7, H8, H9
29 (d) (ii)	2	9.6.4.2.7	H14
29 (d) (iii)	4	9.6.4.3.4	H9, H10, H14
Section II			
Question 30 — Astrophysics			
30 (a) (i)	2	9.7.3.1, 9.7.3.2.3, 9.7.3.2.4, 9.7.3.2.5	H7, H8, H14
30 (a) (ii)	2	9.7.3.1, 9.7.3.2.3, 9.7.3.2.4, 9.7.3.2.5	H7, H8, H13, H14

Question	Marks	Content	Syllabus outcomes
30 (b) (i)	2	9.7.2.2.1	H6, H13
30 (b) (ii)	4	9.7.4.1	H8, H12, H13
30 (c)	7	9.7.6	H3, H7, H14
30 (d) (i)	2	9.7.5.1, 9.7.5.2, 9.7.5.4	H6, H7, H14
30 (d) (ii)	3	9.7.5.1, 9.7.5.2, 9.7.5.4	H6, H7, H14
30 (d) (iii)	3	9.7.5.1, 9.7.5.2, 9.7.5.4	H6, H7, H12
Section II			
Question 31 — From Quanta to Quarks			
31 (a) (i)	2	9.8.3.3.1	H7, H10, H14
31 (a) (ii)	2	9.8.3.3.1	H7, H10, H14
31 (b) (i)	1	9.8.3.2.4	H10
31 (b) (ii)	2	9.8.3.2.4	H10
31 (b) (iii)	3	9.8.3.2.6	H6, H7, H10, H14
31 (c)	7	9.8.4	H3, H14
31 (d) (i)	2	9.8.1.2.3, 9.8.1.3.3	H7, H8, H10, H12
31 (d) (ii)	3	9.8.2.2.1, 9.8.2.2.4, 9.8.2.3.1	H7, H8, H10, H12
31 (d) (iii)	3	9.8.2.2.1, 9.8.2.2.4, 9.8.2.3.1	H7, H8, H10, H14
Section II			
Question 32 — The Age of Silicon			
32 (a) (i)	2	9.9.5.2.1, 9.9.5.3.1, 9.9.5.3.2	H13
32 (a) (ii)	2	9.9.5.2.2	H13, H14
32 (b) (i)	4	9.9.6.2.5, 9.9.6.3.3, 9.9.6.3.1, 9.9.6.3.2	H13
32 (b) (ii)	2	9.9.6.3.1, 9.9.6.3.2	H13, H14
32 (c)	7	9.9.1.2.3, 9.9.1.3.1, 9.9.7.1	H3, H7
32 (d) (i)	2	9.9.2.2.6, 9.9.3.2.1	H7, H14
32 (d) (ii)	3	9.9.3.2.2, 9.9.2.2.5, 9.9.2.3.3	H12, H14
32 (d) (iii)	3	9.9.3.2.2, 9.9.2.3.3, 9.9.5.2.1	H12, H14

2005 HSC Physics Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H12, H6, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly substitutes into relevant formula	2
• Rearranges Kepler's Law of Periods into a ratio of periods OR • Uses the equation from the data sheet and substitutes incorrectly	1

Question 16 (b)

Outcomes assessed: H12, H6, H13

MARKING GUIDELINES

Criteria	Marks
• Correct substitution	3
• Manipulates the equations correctly	2
• Identifies relevant equation	1

Question 17 (a)

Outcomes assessed: H2, H6, H14

MARKING GUIDELINES

Criteria	Marks
Correctly states one prediction	1

Question 17 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Identifies and provides features and characteristics of the correct experiment	2
Identifies the correct experiment	1

Question 17 (c)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Names and describes two advances and relates them to the experiment	3
- Names two advances OR - Names and describes one advance and relates it to the experiment	2
Names one advance	1

Question 18

Outcomes assessed: H1, H2, H6, H8, H14

MARKING GUIDELINES

Criteria	Marks
Makes a judgement of the impact of the result based on evidence provided	3–4
Relates the results to Einstein's Special Theory of Relativity	2
States the result of the Michelson and Morley experiment	1

Question 19

Outcomes assessed: H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
Presents a logical, coherent argument to support Jim Lovell's statement, for the period after the rocket burn including correct, relevant description of Newton's laws	4
Describes and explains the relevance of Newton's laws to the solution of the astronaut's problem for the period after the rocket burn	3
Correctly identifies and relates one of Newton's laws to the situation for the periods either before or after the rocket burn OR	2
States two of Newton's laws	
Correctly states one of Newton's laws	1

Question 20

Outcomes assessed: H7, H9, H12

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of the methods used to assess reliability of information from first hand data or secondary sources - Identifies at least two sources of information - Links methods of assessing the sources of information identified - Describes two applications of induction	5–6
- Outlines methods of assessing reliability of information from first hand or secondary sources - Identifies at least one source of reliable information - Outlines two applications of induction	3–4
- States one method of assessing reliability of information - States one application of induction	1–2

Question 21 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Indicates that there is a repulsive force between the rods AND	2
Provides reason why the conductor jumps upwards	
Provides either of the points above	1

Question 21 (b)

Outcomes assessed: H12, H9, H13

MARKING GUIDELINES

Criteria	Marks
• Correct substitution into correct equations	3
• Correctly relates the two equations	2
• Identifies that magnetic and gravitational force are necessary	1

Question 21 (c)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
• Identifies any correct implication	1

Answers could include:

Cables must be far enough apart.

Cables can be designed to minimise the effect of the force.

Question 22 (a)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies formula and substitutes correctly	1

Question 22 (b)

Outcomes assessed: H4, H7, H9

MARKING GUIDELINES

Criteria	Marks
• Identifies eddy current loss in transformer AND • Identifies resistive losses in transmission line	2
• Identifies eddy current loss in transformer OR • Identifies resistive losses in transmission line	1

Question 22 (c)

Outcomes assessed: H7, H9

MARKING GUIDELINES

Criteria	Marks
• Links superconductivity to zero resistance in eliminating power loss	2
• Identifies what superconductivity means	1

Question 23

Outcomes assessed: H1, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Describes black body radiation and clearly relates this to the resultant change in the direction of scientific thinking	3
• Outlines black body radiation and relates this correctly to the change in direction of scientific thinking	2
• Outlines the correct change in direction of scientific thinking OR • Outlines black body radiation	1

Question 24

Outcomes assessed: H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Provides at least two diagrams with text showing understanding of BCS theory in terms of electron pairs • Correctly identifies role of phonon	4
• Provides two diagrams with text showing understanding of BCS theory in terms of electron pairs	3
• Provides at least one diagram with relevant text showing distortion of lattice	2
• Provides a single labelled diagram of lattice or text to show understanding of concept	1

Question 25 (a)

Outcomes assessed: H7, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly draws line	1

Question 25 (b)

Outcomes assessed: H7, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Provides correct answer	1

Question 25 (c)

Outcomes assessed: H7, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Must cross x axis at 1.6 eV (or where the first line crossed the axis) - Slope greater than in part (a) but not negative	2
- Must cross x axis at 1.6 eV (or where the first line crossed the axis) OR - Slope greater than in part (a) but not negative	1

Question 25 (d)

Outcomes assessed: H7, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly relates two aspects of their new graph to intensity, number of electrons and minimum energy required (work function)	2
Correctly relates one aspect of their new graph to intensity, number of electrons or minimum energy required (work function)	1

Question 26 (a)

Outcomes assessed: H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly draws electric field lines indicating direction	1

Question 26 (b)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
Correctly substitutes in equation	1

Question 26 (c)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Correctly relates electrostatic force to weight force, correctly substitutes and correctly identifies sign	3
Any two of the above	2
Any one of the above	1

Question 27 (a)

Outcomes assessed: H6, H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Equates the centripetal force to the magnetic force	2
• Identifies one appropriate equation	1

Question 27 (b)

Outcomes assessed: H6, H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates mass	2
• States the relationship OR • Uses mass dilation formula	1

Question 27 (c)

Outcomes assessed: H6, H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correct substitution in the equation	2
• Correctly identifies the Lorentz transformation	1

Section II

Question 28 (a) (i)

Outcomes assessed: H7, H11

MARKING GUIDELINES

Criteria	Marks
• Identifies both appropriate pieces of equipment used	2
• Identifies an appropriate piece of equipment	1

Question 28 (a) (ii)

Outcomes assessed: H7, H14

MARKING GUIDELINES

Criteria	Marks
• Provides characteristics and features of at least two applications of remote sensing of surface types	2
• States one application	1

Question 28 (b) (i)

Outcomes assessed: H8, H12

MARKING GUIDELINES

Criteria	Marks
• Correct calculation of distance	2
• Statement of speed of sound OR • Extra travel time to Tennant Creek	1

Question 28 (b) (ii)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines structure and function of equipment	4
• Indicates how data is analysed	
• Outlines structure AND function of equipment	3
• Outlines structure OR function of equipment	2
• States one technique	1

Question 28 (c)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
- Describes current technologies used in measuring Earth's changing magnetic field - Describes principle of plate tectonics - Makes a judgement about the effectiveness of these technologies in providing supporting evidence	6–7
- Describes current technologies used in measuring Earth's changing magnetic field - Describes principle of plate tectonics	4–5
- Outlines a difference between the earth's magnetic field now and in the past - Outlines one area of magnetic evidence for continental drift	2–3
States a feature of the earth's magnetic field that changes over time	1

Question 28 (d) (i)

Outcomes assessed: H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
Describes the effects of extra continental mass on the pendulum	2
Identifies increased gravitational attraction	1

Question 28 (d) (ii)

Outcomes assessed: H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
Explains how the lower density and less deflection explain the possibility of the continents floating	3
Outlines why the lower density crust extending deeper deflects the pendulum less	2
States lower density crust extending deeper	1

Question 28 (d) (iii)

Outcomes assessed: H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of variation of period of pendulum across the world - Describes the shape of the earth as a consequence	2–3
- States variation of period of pendulum OR - Describes the shape of the earth	1

Question 29 (a) (i)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
Clearly shows relationship between injection of radioactive isotopes into bloodstream making blocked and cleared arteries visible and how the decay products of radioactivity are detected	2
- Outlines how radioactive decay products are detected using sensors OR - States blocked artery in ‘before’ scan has been cleared and is visible in ‘after’ scan	1

Question 29 (a) (ii)

Outcomes assessed: H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the radioisotope and provides one reason for the choice	2
- Correctly identifies the radioisotope OR - Provides one justification for incorrect choice	1

Question 29 (b) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Substitutes correctly into equation (with or without conversion to a percentage)	2
• Correctly identifies the equation which needs to be used	1

Question 29 (b) (ii)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
• States more than one physics principle in each and compares	3–4
• States one physics principle in each and compares	2
• States one physics principle in either x-ray or endoscope	1

Question 29 (c)

Outcomes assessed: H3, H5, H14

MARKING GUIDELINES

Criteria	Marks
• Describes technological advances in ultrasound imaging • Identifies current issues and provides arguments for and/or against the use of this technology (discusses current issue)	6–7
• Describes two or more technological advances and discusses one issue OR • Describes one technological advance and discusses two or more current issues	4–5
• Outlines one technological advance in ultrasound imaging • Outlines one current issue	2–3
• Identifies one current issue OR • Makes a correct statement about ultrasound technology	1

Question 29 (d) (i)

Outcomes assessed: H7, H8, H9

MARKING GUIDELINES

Criteria	Marks
- States the purpose of the superconducting magnetic assembly - States the purpose of the RF coils	2
- States the purpose of the superconducting magnetic assembly OR - States the purpose of the RF coils	1

Question 29 (d) (ii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies why gadolinium has been introduced AND - Correctly describes the effect it has had	2
- Correctly identifies why gadolinium has been introduced OR - Correctly identifies what effect gadolinium has had	1

Question 29 (d) (iii)

Outcomes assessed: H9, H10, H14

MARKING GUIDELINES

Criteria	Marks
Identifies at least two advantages of MRI over CAT scans, particularly in reference to the brain	3–4
Identifies two advantages of MRI	2
Identifies one advantage of MRI over CAT	1

Question 30 (a) (i)

Outcomes assessed: H7, H8, H14

MARKING GUIDELINES

Criteria	Marks
• States two differences	2
• States one difference	1

Question 30 (a) (ii)

Outcomes assessed: H7, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies motion of both stars	2
• Identifies motion of one star	1

Question 30 (b) (i)

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly substitutes into appropriate formula	2
• Correctly measures from diagram	1

Question 30 (b) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Calculates a value for the ratio of apparent brightness	4
• Correctly identifies formula to obtain ratio of apparent brightness from two apparent magnitudes	3
• Calculates two values for apparent magnitudes OR • Substitutes absolute magnitudes into formula	2
• Correctly identifies formula to convert absolute magnitude to apparent magnitude OR • Correctly identifies formula to obtain ratio	1

Question 30 (c)

Outcomes assessed: H3, H7, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of astrophysical data collection technology - Identifies evolution of stars - Explains link between improved technology and data collection and stellar evolution	7
- Describes astrophysical data collection techniques - Identifies an idea of stellar evolution	5–6
Describes at least two astrophysical data collection techniques	3–4
Identifies astrophysical data collection techniques	1–2

Question 30 (d) (i)

Outcomes assessed: H6, H7, H14

MARKING GUIDELINES

Criteria	Marks
- Relates period of variable to absolute magnitude - Relates apparent magnitude to absolute magnitude to calculate distance	2
One of the above	1

Question 30 (d) (ii)

Outcomes assessed: H6, H7, H14

MARKING GUIDELINES

Criteria	Marks
Gives reasons for the cause of each of the features <i>A</i>, <i>B</i> and <i>C</i>	3
Gives reasons for the cause of two of the features <i>A</i>, <i>B</i> and <i>C</i>	2
- Correctly identifies that brighter star of pair is eclipsed at <i>A</i> OR - Correctly identifies that dimmer star of pair is eclipsed at <i>B</i> OR - Gives a reason for <i>C</i> OR - Recognise the eclipsing	1

Question 30 (d) (iii)

Outcomes assessed: H6, H7, H12

MARKING GUIDELINES

Criteria	Marks
• Substitutes correctly and evaluates	3
• Obtains period of rotation from graph	2
• Correctly identifies the appropriate formula	1

Question 31 (a) (i)

Outcomes assessed: H7, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Relates observed features of the tracks to properties of α -particle	2
• Makes a correct statement about alpha particles	1

Question 31 (a) (ii)

Outcomes assessed: H7, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies β or γ radiation AND • Identifies features and characteristics of correct tracks to highlight differences	2
• Identifies β or γ radiation	1

Question 31 (b) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies mass number	1

Question 31 (b) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Correctly identifies nuclide and mass number	2
- Correctly names nuclide OR - Correctly states mass number	1

Question 31 (b) (iii)

Outcomes assessed: H6, H7, H10, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of Pauli's contribution - Outlines Fermi's explanation of beta decay	3
- Outlines Pauli's contribution OR - Outlines Fermi's explanation	2
- States KE not conserved OR - States Pauli's suggested particle	1

Question 31 (c)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of at least two technologies stemming from the Manhattan Project - Makes a judgement about the significance of the technologies supported by relevant arguments	6–7
- Provides features and characteristics of at least one technology stemming from the Manhattan Project - Outlines the significance of technology	4–5
- Outlines a technology stemming from the Manhattan Project - Identifies the significance of a technology	2–3
Identifies a technology stemming from the Manhattan Project	1

Question 31 (d) (i)

Outcomes assessed: H7, H8, H10, H12

MARKING GUIDELINES

Criteria	Marks
• Correct substitution in correct formula	2
• Identifies correct formula	1

Question 31 (d) (ii)

Outcomes assessed: H7, H8, H10, H12

MARKING GUIDELINES

Criteria	Marks
• States de Broglie's hypothesis • Correct substitution in correct formula	3
• States de Broglie's hypothesis AND • Identifies correct formula OR • Correct substitution in correct formula	2
• States de Broglie's hypothesis OR • Identifies correct formula	1

Question 31 (d) (iii)

Outcomes assessed: H7, H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Provides characteristics and features of de Broglie's hypothesis and gives reasons why it supports the postulates of Bohr's Theory	3
• Outlines de Broglie and Bohr's ideas	2
• Identifies the postulates of either de Broglie or Bohr	1

Question 32 (a) (i)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
• Correctly evaluates $\overline{A \text{ and } B}$	2
• Correctly evaluates $A \text{ and } B$	1

Question 32 (a) (ii)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
- Provides features and characteristics of a ‘half-adder’ AND - Draws correct circuit diagram with logic gates	2
- Constructs a truth table for the ‘half adder’ OR - Outlines term ‘half adder’ OR - Draws a circuit diagram	1

Question 32 (b) (i)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
- Draws correct diagram - Provides features and characteristics of open-loop and closed-loop gain - Calculates correct values for R_1, R_2	4
- Draws correct diagram - Outline of open-loop OR closed-loop gain - Correct values for R_1, R_2	2–3
Draws basic circuit for inverting amplifier	1

Question 32 (b) (ii)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
Indicates on diagram that output is triangular wave truncated at -15 V	2
- States that output is amplified truncated wave OR - Draws appropriate diagram	1

Question 32 (c)

Outcomes assessed: H3, H7

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of advances in at least two semiconductor technologies and links the advances in technology to new applications	7
Describes a number of changes in technology and describes applications and makes some attempt to link them	6
Outlines a number of changes in technology applications	4–5
- Identifies technology changes AND/OR - Identifies applications	1–3

Question 32 (d) (i)

Outcomes assessed: H7, H14

MARKING GUIDELINES

Criteria	Marks
Distinguishes difference between input and output transducers, giving examples of each	2
Provides examples of one input and one output transducer	1

Question 32 (d) (ii)

Outcomes assessed: H14, H12

MARKING GUIDELINES

Criteria	Marks
• Correct calculation of R	3
• Identification of currents through both resistors, using correct calculation	2
• Correct use of $V = IR$	1

Question 32 (d) (iii)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates voltage at input of inverter and hence calculates 1 or 0 output	3
• Correct use of $V = IR$ to calculate voltages at input of inverter OR • Correct use of not gate	2
• Correct identification of LDR resistances from table OR • Indicates input voltage required for G to give output of 1 or 0	1